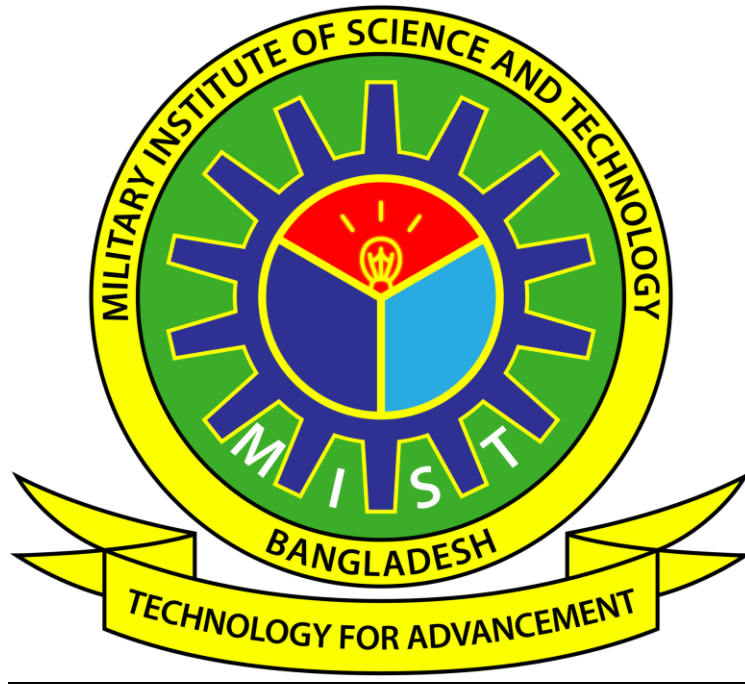




Military Institute of Science and Technology

Department of Electrical Electronic and Communication
Engineering

Project Name	<u>Open Ended Lab Project</u>
Course Title	Power System Laboratory-I
Course code	EECE-306

GROUP-02

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Task 1: Perform load flow studies of the power system of Fig. 1 to identify slack bus (Bus no. 1) power and bus voltages (Bus no. 2 to Bus no. 3). Compute line flows and line losses also. [Use ETAP simulation software]

Solution: To perform this load flow analysis we used ETAP Simulation software. Here are the results after simulation of the given 5 bus system. We have identified slack bus (Bus no. 1) power and bus voltages (Bus no. 2 to Bus no. 3). We have also computed the line flows and line losses also using Etap software.

In the load flow analysis by ETAP, we used Adaptive Newton Raphson Method with maximum 99 number of iterations and precision 0.001. All the loads have been assumed as 100% motor. Ratings of generators, loads, impedances has given according to question. We used typical data of ETAP for generator 1, 2.

One-Line Diagram - OLV1 (Load Flow Analysis)

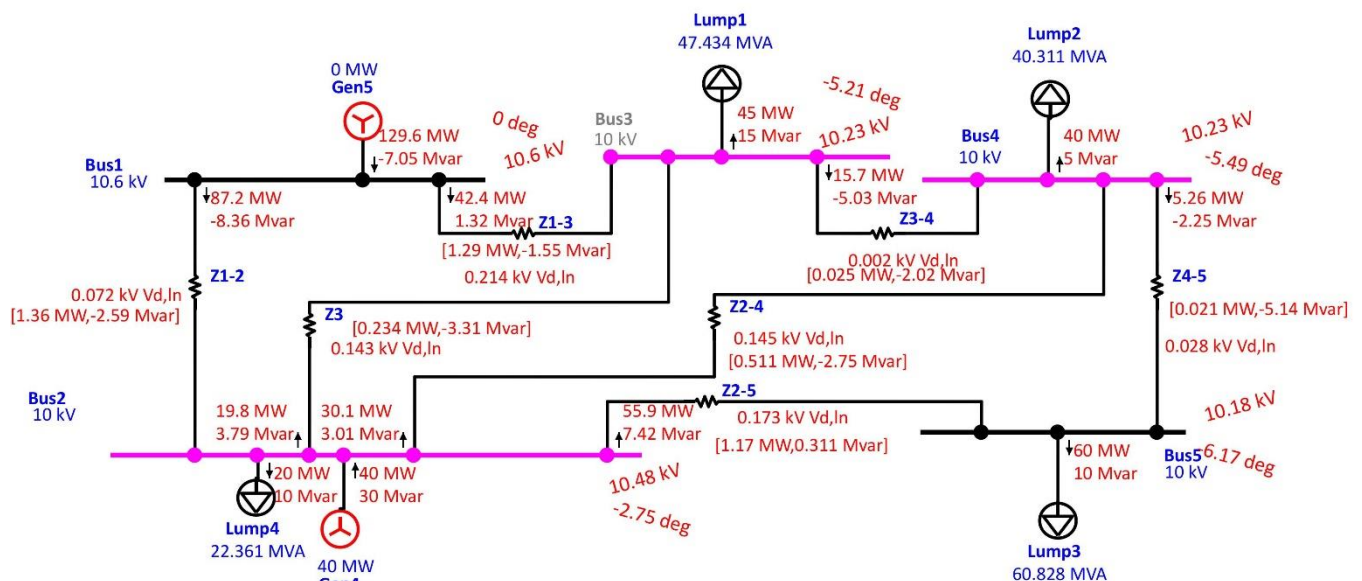


Fig-Load Flow Analysis

LOAD FLOW REPORT

Bus		Voltage		Generation		Load		Load Flow					XFMR
ID	kV	% Mag.	Ang.	MW	Mvar	MW	Mvar	ID	MW	Mvar	Amp	%PF	%Tap
* Bus1	10.600	100.000	0.0	129.612	-7.048	0.000	0.000	Bus2	87.223	-8.364	4772.6	-99.5	
								Bus3	42.389	1.316	2309.9	100.0	
Bus2	10.000	104.758	-2.8	40.000	30.000	20.000	10.000	Bus1	-85.864	5.777	4742.9	-99.8	
								Bus4	30.132	3.011	1668.9	99.5	
								Bus5	55.930	7.422	3109.5	99.1	
								Bus3	19.802	3.791	1111.2	98.2	
Bus3	10.000	102.289	-5.2	0.000	0.000	45.000	15.000	Bus1	-41.097	-2.867	2325.3	99.8	
								Bus2	-19.568	-7.103	1175.0	94.0	
								Bus4	15.666	-5.030	928.7	-95.2	
Bus4	10.000	102.254	-5.5	0.000	0.000	40.000	5.000	Bus2	-29.621	-5.763	1703.8	98.2	
								Bus3	-15.641	3.013	899.3	-98.2	
								Bus5	5.261	-2.250	323.1	-91.9	
Bus5	10.000	101.764	-6.2	0.000	0.000	60.000	10.000	Bus2	-54.760	-7.111	3132.8	99.2	
								Bus4	-5.240	-2.889	339.5	87.6	

* Indicates a voltage regulated bus (voltage controlled or swing type machine connected to it)
Indicates a bus with a load mismatch of more than 0.1 MVA

Branch Losses Summary Report

Branch ID	From-To Bus Flow		To-From Bus Flow		Losses		% Bus Voltage		Vd % Drop in Vmag
	MW	Mvar	MW	Mvar	kW	kvar	From	To	
Z1-2	87.223	-8.364	-85.864	5.777	1358.6	-2587.2	100.0	104.8	1.17
Z1-3	42.389	1.316	-41.097	-2.867	1291.4	-1550.4	100.0	102.3	3.71
Z2-4	30.132	3.011	-29.621	-5.763	511.2	-2752.4	104.8	102.3	2.50
Z2-5	55.930	7.422	-54.760	-7.111	1170.2	311.0	104.8	101.8	2.99
Z3	19.802	3.791	-19.568	-7.103	234.0	-3312.5	104.8	102.3	2.47
Z3-4	15.666	-5.030	-15.641	3.013	25.0	-2017.0	102.3	102.3	0.04
Z4-5	5.261	-2.250	-5.240	-2.889	21.3	-5139.1	102.3	101.8	0.49
					4611.7	-17047.7			

* This Transmission Line includes Series Capacitor.

Table 1:Load flow analysis

Bus No	Generation		Load		Bus voltage	
	MW	MVAR	MW	MVAR	Voltage(p.u)	Phase angle(°)
1	129.612	-7.048	0	0	1.06	0
2	40	30	20	10	1	-2.8
3	0	0	45	15	1	-5.2
4	0	0	40	5	1	-5.5
5	0	0	60	10	1	-6.2

Task 2: Verify the results obtained in Task no. 1 by writing a MATLAB code adopting any load flow analysis method.

Solution:Here we have to verify our results using any load flow analysis method in MATLAB software.

We have used Gauss Seidal Iterative Method in this case.Here's our MATLAB code-

```
%group2
```

```
clc
clear all
close all
```

```
j          = sqrt(-1)
bus_number=5;
iter       = 1000
```

```
%Bus Data
```

```
%      busNO      BusVoltage      Pg      Qg      Pl      Ql
bus_data = [ 1      1.06  0.0      0      0      0      0
              2      1.0  0.0      40     30     20     10
              3      1.0  0.0      0      0      45     15
              4      1.0  0.0      0      0      40      5
              5      1.0  0.0      0      0      60     10 ]
```

```
%transmission line Data
```

```
%      start end      R      X      Line charging(B/2)
line_data = [1      2      0.02      0.06      0  0.015
              1      3      0.08      0.24      0  0.0125
              2      3      0.06      0.25      0  0.01
              2      4      0.06      0.18      0  0.01
              2      5      0.04      0.12      0  0.0075]
```

3	4	0.01	0.03	0	0.005
4	5	0.08	0.24	0	0.0125]

```

%formation of admittance matrix
R      = line_data(:,3); %resistance between the buses
X      = line_data(:,4); %reactance between buses
sus     = j*line_data(:,6);
line_n  = length(line_data(:,1)); %number of lines
Z      = R+j*X;           %impedance
Y      = ones(line_n,1)./Z; %admittance
Yt     = zeros(5,5);

bus_n   = length(bus_data(:,1)); %number of buses

%upper and lower triangular matrix
for i = 1:line_n
    Yt(line_data(i,1),line_data(i,2)) = - Y(i);
    Yt(line_data(i,2),line_data(i,1)) = Yt(line_data(i,1),line_data(i,2));
end
%principal diagonal elements
for i=1:bus_n
    for k=1:line_n
        if line_data(k,1) == i
            Yt(i,i) = Yt(i,i)+Y(k)+sus(k);
        elseif line_data(k,2) == i
            Yt(i,i) = Yt(i,i)+Y(k)+sus(k);
        else
            end
        end
    end
end

Yt

%Voltage and Phase of Swing, PV and PQ buses
%voltage
baseMVA = 100;
bus_v   = bus_data(:,2); %pu bus voltage
phase_b = bus_data(:,3); %phase of buses
P       = bus_data(:,4)-bus_data(:,6);
Q       = bus_data(:,5)-bus_data(:,7);
V       = zeros(bus_n,1);
V       = bus_v;         %initial assumptions

V_new   = V;
for iterations =1:iter
    for i=2:bus_n
        Sum = 0;
        for k = 1:bus_n
            if i~=k
                Sum = Sum + Yt(i,k)*V_new(k);
            end
        end
        V(i) = (((P(i)-1j*Q(i))/baseMVA)/conj(V_new(i)))-Sum)/Yt(i,i);
        V_new(i) = V(i);
    end
end

```

```

end
V_new;
V;

%for imaginary part

img      = zeros(bus_n,1);
for i=1:bus_n
    img(i,1) = imag(V(i,1));
end
img;

%for real part

real_part    = zeros(bus_n,1);
for re=1:bus_n
    real_part(re,1) = real(V(re,1));
end
real_part;
%angle
angle = zeros(bus_n,1);
for Angle=1:bus_n
    angle(Angle,1) = atan2(img(Angle,1)/real_part(Angle,1));
end

magnitude = abs(V);

for i=1:bus_n
    fprintf('The voltage of BUS %d is %f \n ',i,magnitude(i,1))
end
fprintf('\n \n \n')
for i=1:bus_n
    fprintf('The phase of BUS %d is %f \n ',i,angle(i,1))
end
%Real and reactive power of Swing Bus
apparent_power = 0;
swing_bus = 1;
sum = 0;
for q =2:bus_n
    sum = sum + Yt(swing_bus,q)*V(q,1);
end

apparent_power=V(swing_bus,1)*(V(swing_bus,1)*Yt(swing_bus,swing_bus)+sum);
P_Swing= real(apparent_power)*100;

fprintf('\nReal power of swing bus is P= %d MW\n' ,P_Swing)

Q_Swing = imag(apparent_power)*100;

fprintf('Reactive power of swing bus is Q= %d MVAR\n' ,Q_Swing)

```


Task 3: Make an undervoltage event in Bus-4 and apply any technique to overcome the undervoltage problem of the system. (Consider, below 85% to be undervoltage for any bus) .

Solution: Undervoltage occurs when the average voltage of a three-phase power system drops below intended levels, and is sometimes referred to as a brown-out. Undervoltage situation can be created by overloading the specific bus.

An undervoltage problem at Bus 4 created by adding a lump load of 10MW,130Mvar.

By this we get a bus voltage of 8.63KV,Which is around 85% of previous voltage.

To solve this undervoltage problem,we then added a Capacitor Bank with 10kv,130Mvar rating.After this we get bus voltage of 10.18Kv.Which is same before the undervoltage problem.

One-Line Diagram - OLV1 (Load Flow Analysis)

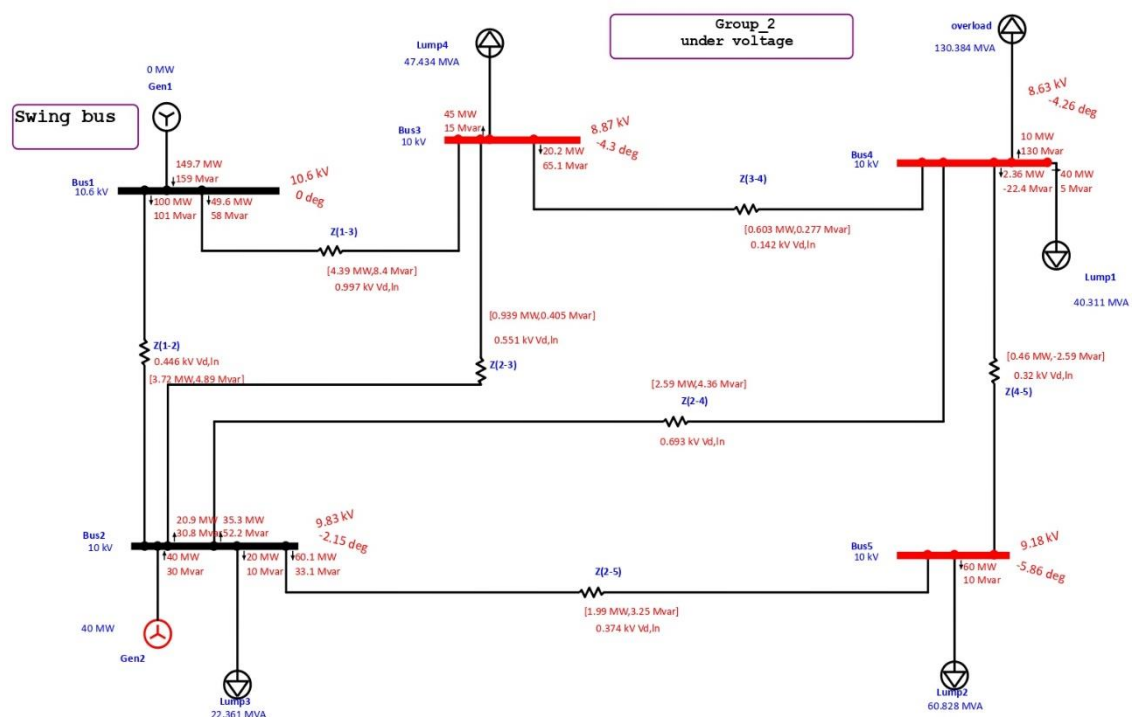


Fig-Undervoltage Condition

One-Line Diagram - OLV1 (Load Flow Analysis)

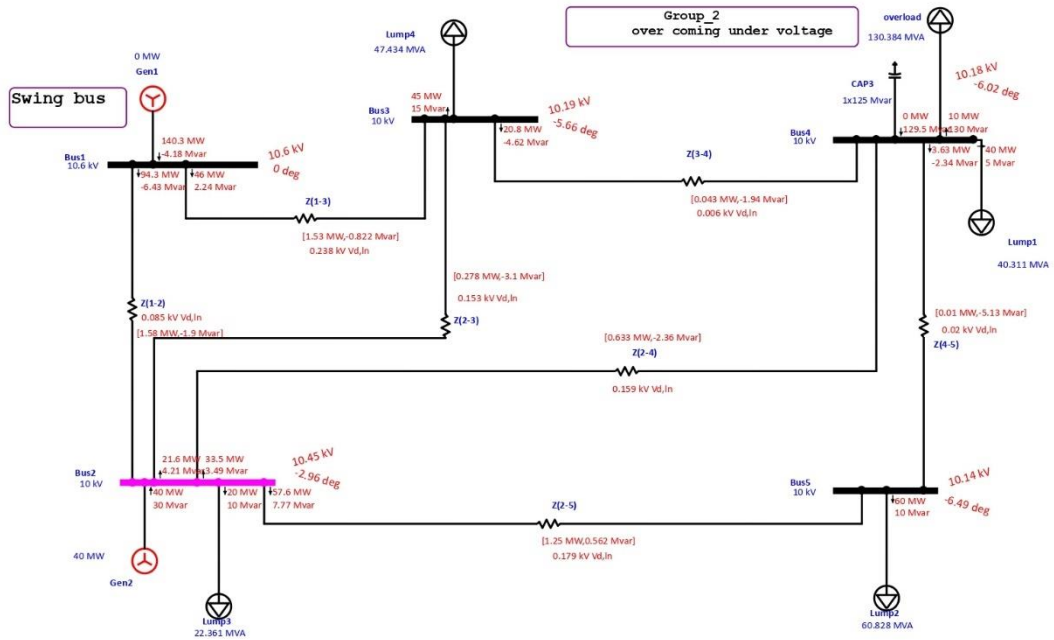


Fig-Undervoltage solved

Task 4: Initiate an unsymmetrical fault (L-G, L-L-G, L-L) in the transmission lines. Group-1 initiate fault in line 1-3, Group-2 initiate fault in line 2-3, Group-3 initiate fault in line 2-4 and Group-4 initiate fault in line 2-5. Observe the change in the results of the load flow studies after the fault initiation.

Solution: To initiate unsymmetrical fault(L-L, L-G, L-L-G) in line 2-3, Short Circuit analysis has been selected and then Bus 2 and 3 three were selected from study case. Different types of faults were selected from display options.

Here is the result of short circuit analysis in ETAP software-

One-Line Diagram - OLV1 (Short-Circuit Analysis)

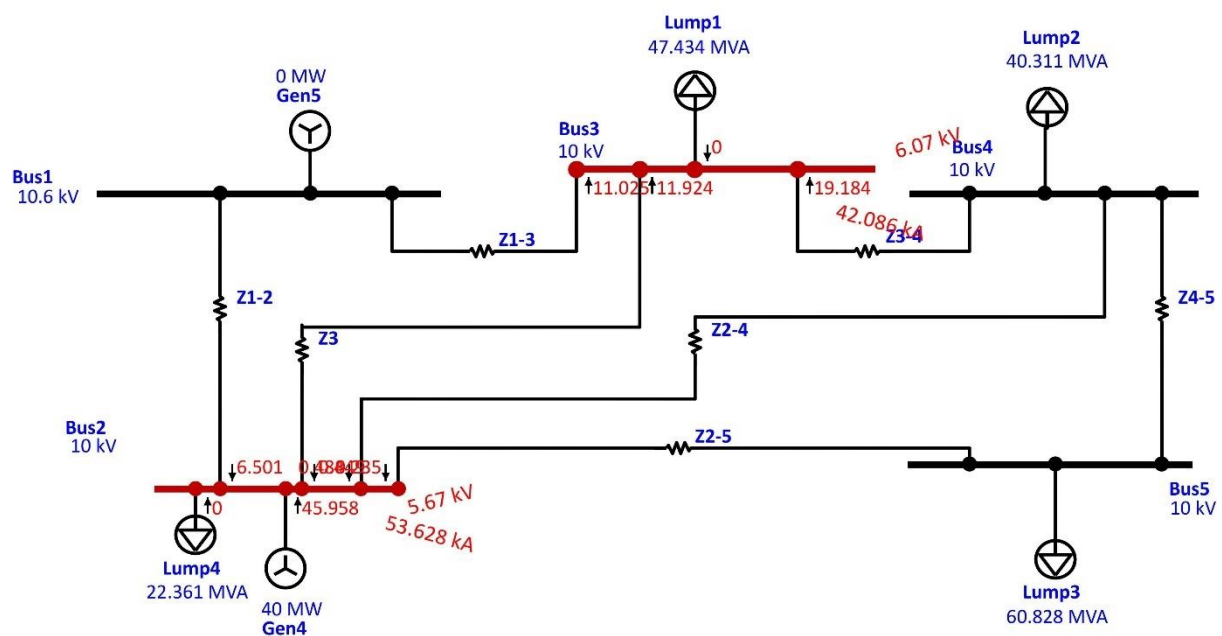


Fig-Unsymmetrical fault(L-G)

One-Line Diagram - OLV1 (Short-Circuit Analysis)

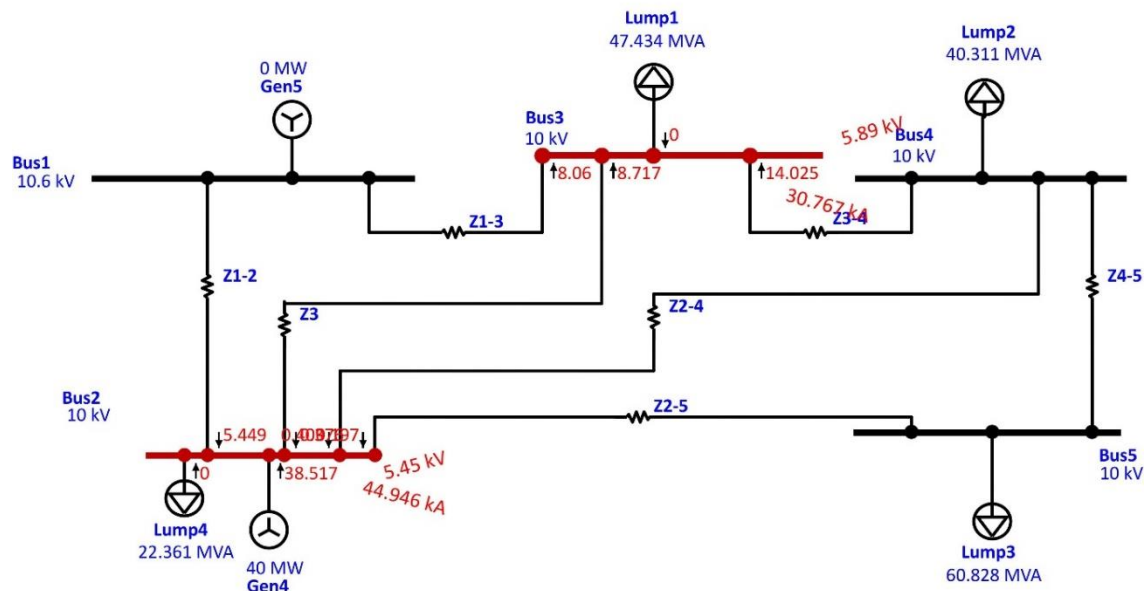


Fig:Unsymmetrical Fault(L-L-G)

One-Line Diagram - OLV1 (Short-Circuit Analysis)

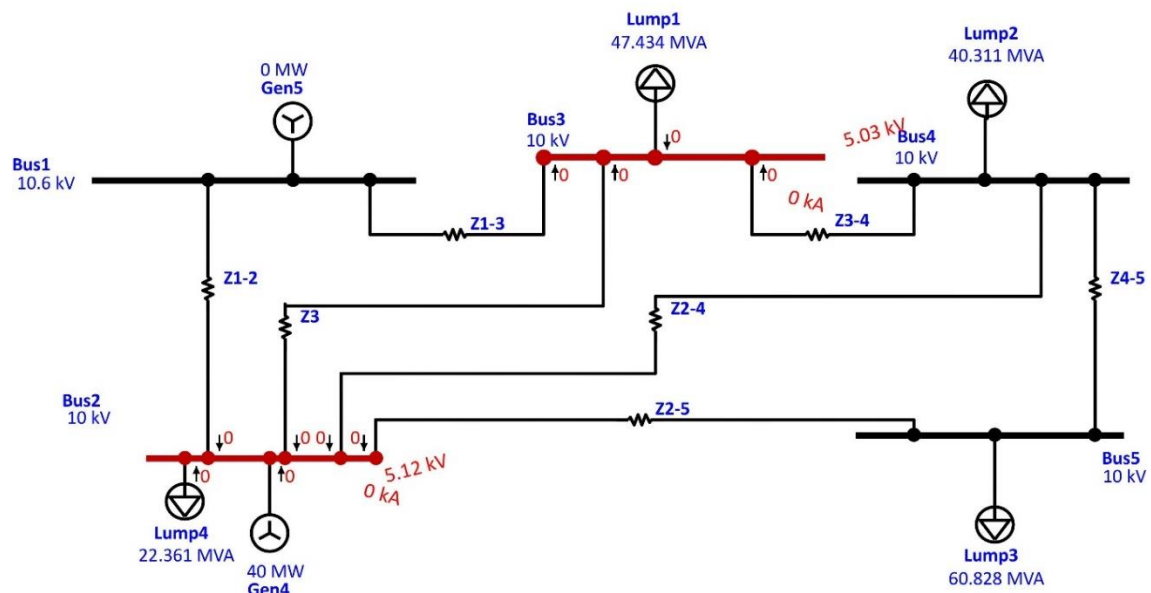


Fig:Unsymmetrical Fault(L-L)

Short-Circuit Summary Report

1.5-4 Cycle - 3-Phase, LG, LL, & LLG Fault Currents

Prefault Voltage = 100 % of the Bus Nominal Voltage

Bus		3-Phase Fault			Line-to-Ground Fault			Line-to-Line Fault			*Line-to-Line-to-Ground		
ID	kV	Real	Imag.	Mag.	Real	Imag.	Mag.	Real	Imag.	Mag.	Real	Imag.	Mag.
Bus2	10.000	4.852	-51.181	51.410	6.750	-53.202	53.628	49.220	5.726	49.552	-50.274	17.077	53.095
Bus3	10.000	5.233	-46.098	46.394	6.769	-41.565	42.113	44.965	5.798	45.337	-46.399	9.731	47.408

All fault currents are symmetrical (1.5-4 Cycle network) values in rms kA.

* LLG fault current is the larger of the two faulted line currents.

Fig: Fault(L-L,L-L-G,L-G) analysis in 2-3 line

Comparison with PowerWorld: This project also simulated in Powerworld and this is the simulation result of load flow analysis in powerworld.

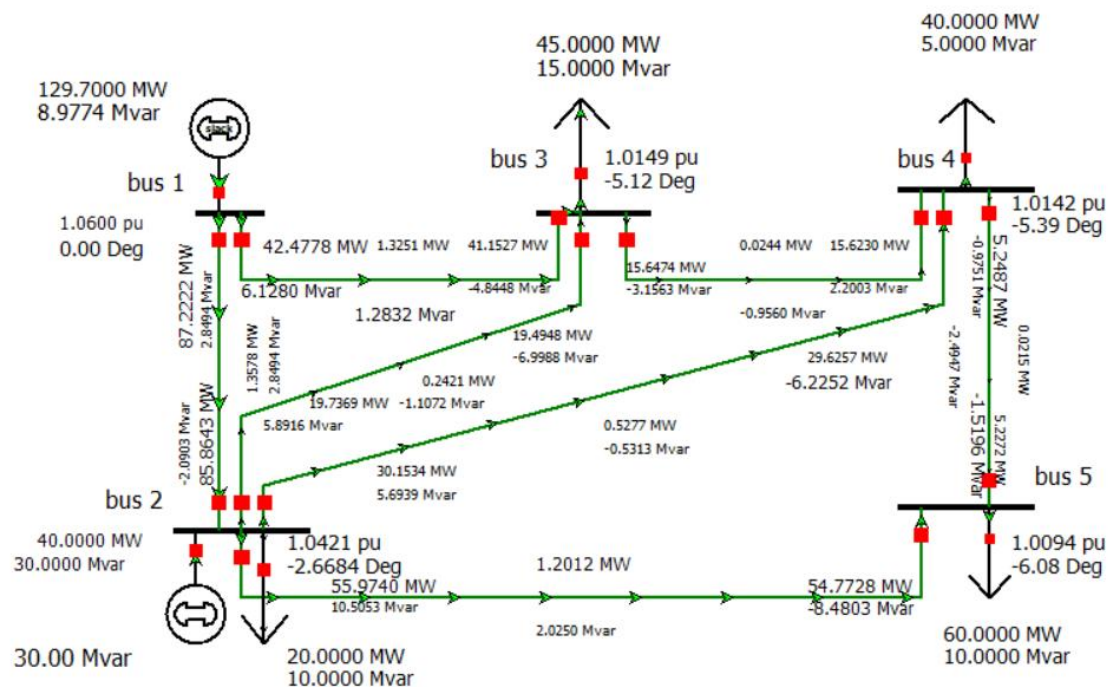


Fig: Load flow analysis with powerworld

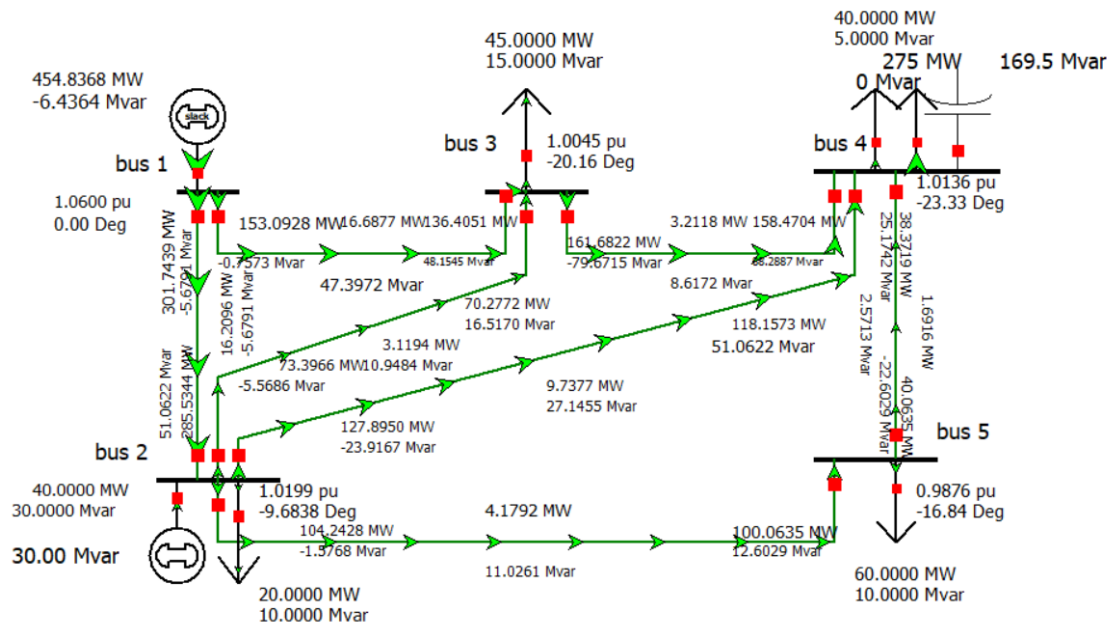


Fig: Undervoltage situation and overcome with powerworld

PowerWorld Fault Analysis Data for Line 2-3:

- Choose the Faulted Line:** 2 (bus 2) [10.00 kV]
- Sort by:** Name (radio button), Number (radio button)
- Search For Near Bus:** 1 (bus 1) [10.00 kV], 2 (bus 2) [10.00 kV], 3 (bus 3) [10.00 kV], 4 (bus 4) [10.00 kV], 5 (bus 5) [10.00 kV]
- Select Far Bus, CKT:** 1 (bus 1) [10.00 kV] CKT 1, 3 (bus 3) [10.00 kV] CKT 1, 4 (bus 4) [10.00 kV] CKT 1, 5 (bus 5) [10.00 kV] CKT 1
- Fault Location:** ☐ Bus Fault, ☒ In-Line Fault
- Fault Type:** ☐ Single Line-to-Ground, ☒ Line-to-Line, ☐ 3 Phase Balanced, ☐ Double Line-to-Ground
- Location %:** 30
- Fault Impedance:** R: 0.00000, X: 0.00000
- Fault Current:** Scale Current By: 1.00000
- If Magnitude:** 12498.800 Amps
- If Scaled Mag:** 12498.800 Amps
- If Angle:** -150.52 deg.
- Units:** ☐ p.u., ☒ Amps
- Subtransient Phase Current:**

	Amps	deg.
A	0.000	0.00
B	12498.500	-150.52
C	12498.500	29.48

Fig: Fault analysis data in line 2-3 in powerworld

The results are almost similar to the findings from ETAP Simulation.

Discussion:

In this project, we conducted power flow analysis using ETAP software and MATLAB to figure out various aspects of an electrical power system. We focused on analysing the characteristics of the swing bus, which helped in understanding how power balance was maintained. We also explored the impact of undervoltage conditions and how they affect system performance. Load flow analysis allowed us to identify voltage levels and power distribution across the network. Furthermore, we studied unsymmetrical faults to examine how the system behaves during unbalanced conditions. The combination of ETAP and MATLAB provided accurate simulations, and the results were used to evaluate the overall system stability and efficiency. We also conducted same process in powerworld simulation software, the results were similar with ETAP simulation and MATLAB.

References:

1. Elements of Power System Analysis, Stevenson.
2. Power System Analysis, Hadi Sadaat.
3. Internet for theoretical references.